

ProjectPlan

LifeLine — Comprehensive Blood Donation Platform

Document: Project Plan

Course: CSC13002 - Introduction to Software Engineering

Team: Sanguine (Team 05)

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Revision History

Date	Version	Description	Author
05/06/2026	1.0	Initial draft – Project Plan (Roles, Responsibilities, Schedule, Build Plan)	Trần Anh Kiệt
12/06/2026	1.1	Initial draft – Project Plan (Introduction, Project Overview, Risk Management)	Trần Anh Kiệt
11/07/2026	1.2	Refined headings, Added due date for tasks	Trần Anh Kiệt
23/07/2026	1.3	2nd revision addressing TA feedback: added Task Review Process (Sec 3.4) with a dedicated reviewer per task; clarified task assignment (split combined/vague owner entries into individually-owned tasks with explicit scope) across Sec 4.2.2–4.2.5; rebalanced Sprint 5 workload by moving some Functional Groups forward into Sprint 4. See Changes.md for the full list.	Trần Anh Kiệt

1. Introduction

This document describes the overall plan for the development of LifeLine, a web-based blood donation management platform, developed by Team Sanguine (Group 05) as part of the CSC13002 Introduction to Software Engineering course. It covers the project organization, sprint schedule, risk management strategy, and build plan to guide the team throughout the development lifecycle.

The project applies the Scrum framework, organized into five sprints each corresponding to one Project Assignment (PA1 through PA5). This document serves as the top-level plan used by the Project Manager to direct the team and track progress. It will be updated incrementally as the project evolves and TA feedback is incorporated in subsequent PAs.

The following people use this Project Plan:

- **The Project Manager** uses it to plan the sprint schedule and resource allocation, and to track progress against milestones.
- **Team members** use it to understand their responsibilities, task deadlines, and dependencies on other team members' work.
- **The Teaching Assistant and course instructors** use it to evaluate the team's planning quality and provide feedback for subsequent PAs.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

The purpose of this project is to develop LifeLine, a modern web-based blood donation management platform that connects donors, blood donation centers, and hospitals within a unified ecosystem. The platform addresses the fragmented and manual nature of current blood donation processes by providing online registration, appointment booking, donation tracking, and emergency blood request notifications.

The scope of the project includes the design and full-stack development of a responsive web application supporting three primary user roles: Donors, Blood Center Staff, and Hospital Staff. The system will also include an AI-powered Q&A assistant for donor guidance, a gamification layer for donor engagement, blood inventory management, and an SOS emergency broadcast system.

The key objectives are:

- To provide a seamless, accessible platform for donors to register, book appointments, track their donation history, and receive personalized support.
- To enable blood donation centers to manage campaigns, donor check-ins via QR code, blood inventory, and donor communications from a centralized interface.
- To allow hospitals to submit SOS blood requests that are instantly broadcast to eligible donors and blood centers via push notifications and email.
- To integrate an AI-powered donor assistant that provides eligibility guidance, pre- and post-donation advice, and answers frequently asked questions.
- To foster long-term donor engagement through gamification features including milestone badges, achievement levels, and progress-tracking visualizations.

The final deliverables will include:

- A fully functional, responsive web application covering all 14 functional groups.
- A backend system supporting donor records, campaign management, blood inventory, SOS broadcasting, and AI assistant integration.
- Complete project documentation including Vision Document, Use-Case Model & Specifications, Software Architecture, Test Plan, and Reflective Report.
- Spec Kit artifacts (constitution.md, generated test cases) for all implemented functional groups.

2.2 Assumptions and Constraints

Key assumptions are:

- Target users (donors, blood center staff, hospital staff, admin) have basic familiarity with web-based applications and smartphones.
- All five team members will remain available and actively contribute throughout the entire project duration.
- The team will receive regular feedback from the Teaching Assistant and instructors to validate alignment with course requirements.

- Third-party services (Maps API, push notification provider, LLM/AI API) will remain accessible and stable throughout development.

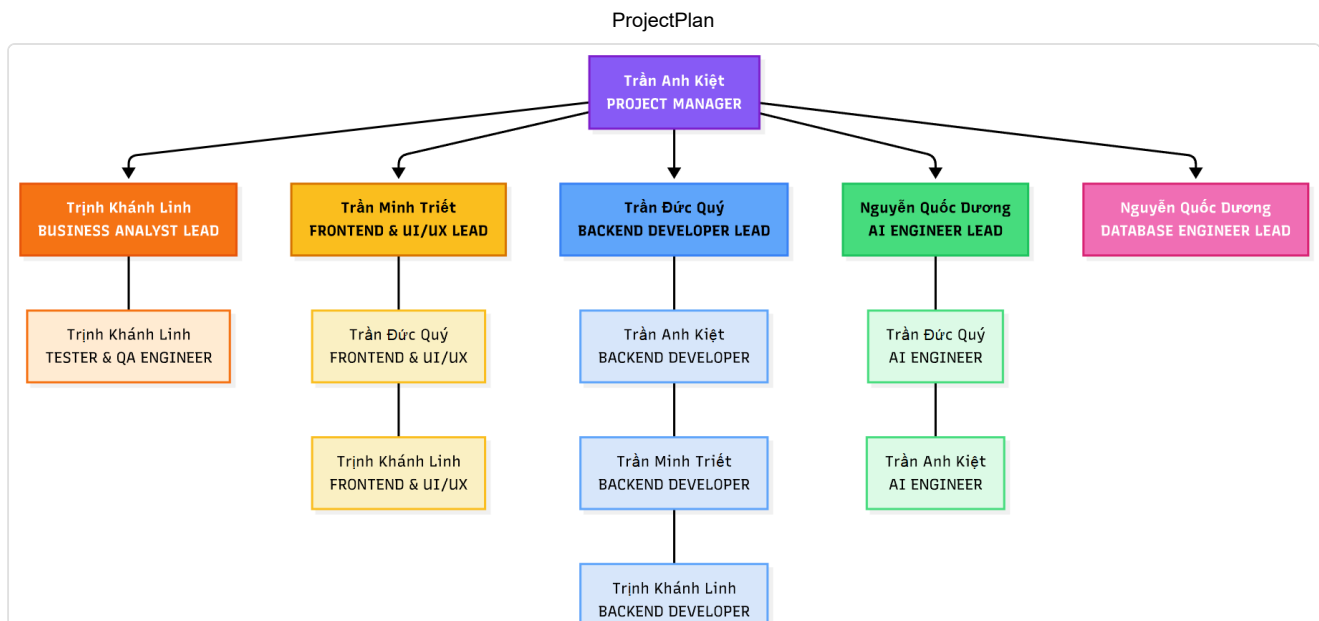
Key constraints are:

- The project has no budget; development will rely exclusively on free-tier and open-source tools.
- The team is small (five members), requiring efficient task distribution and cross-functional collaboration across all layers of the stack.
- The project must be completed within the semester timeline (approximately 13 weeks, May 25 – August 22, 2026), with all five PAs submitted on schedule.
- The platform will be developed and tested primarily on commonly used devices (laptops and smartphones) and will target modern web browsers.

3. Project Organization

This section defines the organizational structure of the Sanguine team for the LifeLine project. All members act as full-stack engineers; the roles below indicate each member's primary leadership area while everyone contributes across the stack.

3.1 Team Structure



3.2 Team Roles and Responsibilities

Member	Primary Role	Responsibilities
Trần Anh Kiệt	Project Manager	Project planning, sprint management, task assignment in Jira, team coordination, meeting organization, progress tracking, communication with TA/instructor, support back-end development.
Trịnh Khánh Linh	Business Analyst	Requirements gathering and analysis, user story creation, use case documentation, stakeholder communication, maintaining project documentation.
Trần Minh Triết, Trần Đức Quý	Frontend Developer & UI/UX Designer	Design wireframes and mockups, create user interfaces, ensure responsive design, improve user experience, implement frontend functionalities and API integration.
Trần Đức Quý, Trần Anh Kiệt	Backend Developer	Design and implement APIs, business logic development, authentication and authorization, server-side processing, integration with database and AI modules.
Nguyễn Quốc Dương, Trần Đức Quý	AI Engineer	Research and implement AI-powered features, train or integrate AI models, optimize AI performance, support intelligent recommendation and chatbot functionalities.
Nguyễn Quốc Dương	Database Engineer	Database design, schema management, query optimization, data consistency, backup and security management.
Trịnh Khánh Linh	Tester & QA Engineer	Unit testing, frontend testing, backend testing, test case creation, bug reporting, quality assurance, validation of system requirements.

Note: All team members are expected to participate in all phases of development. Roles indicate the primary area of expertise and leadership, not exclusivity. All members must register for AI coding accounts (GitHub Copilot for Students, Cursor) per PA1 requirements.

3.3 Risk Management

ID	Risk Description	Probability	Impact	Exposure	Mitigation Strategy
R1	Stakeholder requirements may be incomplete, unclear, or change during development, leading to features that do not fully satisfy user needs.	Medium	High	High	Conduct regular requirement reviews, validate requirements with stakeholders, and maintain up-to-date documentation using Spec-Kit.
R2	Project scope may expand beyond the planned schedule due to additional features or requirement changes.	Medium	High	High	Prioritize core functionalities and defer non-essential features to future releases. Review project scope regularly during sprint planning.
R3	Team members may become unavailable or withdraw during development, affecting project continuity and productivity.	Low	High	Medium	Document all tasks in Jira, perform knowledge sharing, and ensure all members can contribute across multiple project areas.
R4	Team members may require additional time to learn and apply technologies such as React, MongoDB, JavaScript, and Java	Medium	Medium	Medium	Allocate learning time in early sprints, encourage peer support, and utilize available learning resources and development tools.
R5	Integration issues between front-end, back-end, and database components may cause functionality errors or data inconsistencies.	Medium	Medium	Medium	Define API specifications early, perform continuous integration testing, and validate system components throughout development.

ID	Risk Description	Probability	Impact	Exposure	Mitigation Strategy
R6	Sensitive donor information may be exposed due to improper credential management or security vulnerabilities.	Low	High	Medium	Store credentials securely, use environment variables, implement access controls, and follow secure development practices.
R7	Third-party services such as Vercel, Hugging Face, email providers, or external APIs may experience downtime, service limitations, or policy changes.	Medium	High	High	Monitor service availability, maintain backups, and prepare alternative deployment or service options when possible.
R8	System performance may degrade as the number of users, donation records, and blood requests increases.	Low	Medium	Low	Optimize database queries, implement indexing strategies, and perform performance testing before deployment.
R9	Requirements validation may be insufficient, resulting in features that are implemented correctly but do not solve the actual problem.	Medium	High	High	Conduct prototype demonstrations, stakeholder reviews, and user acceptance testing throughout development.

3.4 Task Review Process

To ensure quality and avoid single-person blind spots, every task tracked in Jira is given two distinct people: an **Owner** (does the work) and a **Reviewer** (checks the work). The two roles are never the same person for a given task.

- **Reviewer assignment rule:** For documentation tasks, the reviewer is normally the team's Business Analyst/QA lead (Linh) or the Project Manager (Kiệt) when Linh is the owner. For code/FG implementation tasks, the reviewer is a member who was *not* part of that FG's implementation team, so the review is an independent check rather than self-sign-off.
- **Review workflow in Jira:** (1) Owner moves the task card to "In Review" and links the artifact/PR/doc. (2) Reviewer checks it against the task's stated deliverable/acceptance

criteria and leaves comments directly on the Jira card. (3) If changes are needed, the card moves back to "In Progress"; if approved, the Reviewer moves it to "Done" and it is included in the next Scrum Meeting update. (4) Every task's Owner and Reviewer are now listed explicitly in the per-sprint task tables in Section 4.2.

- **Sprint-level review:** In addition to per-task review, the Sprint Review Meeting at the end of each sprint is where the Reviewer of each major deliverable (UC Specs, Architecture Docs, each implemented FG) reports review outcomes to the whole team before the PA package is finalized.
- **Code review specifics:** For FG implementation tasks, the Reviewer performs a lightweight code review (readability, adherence to the Spec Kit-generated spec, and a manual run-through of the feature) before the branch is merged into `main`.

4. Project Plan

4.1 Project Schedule Overview

The project is organized into 5 Sprints, each corresponding to one Project Assignment (PA). The table below provides the high-level schedule.

Sprint	Period	Corresponding PA	Key Deliverables
1	25 May – 07 Jun 2026	PA1	Project Proposal, App Survey, Team Contract, Tools Setup, Spec Kit Research
2	08 Jun – 21 Jun 2026	PA2	Project Plan, Vision Document (all 14 FGs), Spec Kit Initialization, AI Usage Report
3	22 Jun – 12 Jul 2026	PA3	UC Model & Specs for all 14 FGs, UI Prototypes, Implement FG1 (1.1 User Account Mgmt), v0.1 Alpha
4	13 Jul – 02 Aug 2026	PA4	Architecture Docs (C4 L1–L3, Deployment), Implement FG2 (1.2 Booking) + FG3 (2.1 Campaign Mgmt), v0.2 Beta
5	03 Aug – 22 Aug 2026	PA5	Implement FG4–FG14 (remaining 11 FGs), Test Plan & Execution (≥50 TCs), Reflective Report, Final Demo, v1.0

4.2 Detailed Sprint Plans

4.2.1 Sprint 1 (PA1) – 25 May 2026 to 07 June 2026

Sprint 1 Goal: Complete all PA1 deliverables – Group Registration, Project Proposal, Existing App Survey, Team Contract, and Development Tools Setup. All members also

complete self-learning on Spec Kit.

Item	Task	Details
Sprint Planning	29 May 2026 15:00	Discussed priorities, deadlines, roles.
Scrum Meeting #1	01 Jun 2026 18:00	Reviewed weekend tasks. Delegated detailed feature write-ups for Project Proposal. Assigned mandatory Markdown template for all members.
Scrum Meeting #2	04 Jun 2026 18:30	Reviewed all drafted sections. Assigned final tasks: Sec 1–2 (Linh), Sec 3 review (Dương+Triết), Sec 4 AI refinement (Quý), Sec 5 (Kiệt), Sec 6–10 (Dương), write final proposal (Linh+Quý). Added Spec Kit self-learning task for all.
Sprint Review	06 Jun 2026	Review final PA1 deliverables before submission.

TASK		OUTCOME	WEEK 1 (MAY 25 - 31)							WEEK 2 (JUN 1-7)						
			25/05	26/05	27/05	28/05	29/05	30/05	31/05	01/06	02/06	03/06	04/06	05/06	06/06	07/06
MEETING	Sprint Planning Meeting	Planning report														
	Scrum Meeting #1	Meeting minutes														
	Scrum Meeting #2	Meeting minutes														
	Sprint Review Meeting	Review report														
SETUP	Group Registration & Tools Setup	Jira, GitHub, Zalo, AI accs														
DOCUMENTATION	Outline Survey Report & Existing App Survey	Survey report														
	Final Survey Report	Survey report														
	Team Contract	Contract document														
	Create User Survey Google Form	Google Form														
	Outline Project Proposal	Proposal outline														
	Draft Proposal's Features	Feature write-up														
	Analyze User Survey Responses	Survey analysis														
	1st Advisory Teacher Meeting	Presentation documents														
	Draft Project Proposal	Project Proposal Draft														
	Finalize & compile Project Proposal	Final proposal PDF/MD														
LEARNING	Spec Kit	Spec Kit doc														
	HTML, CSS, JavaScript	Training evidence/screenshots														
	ReactJS, TailwindCSS, NodeJS	Training evidence/screenshots														

Figure 1: Sprint 1 Schedule

Deliverables: PA1 submission zip – Project Proposal (MD+PDF), Existing App Survey (MD+PDF), Team Contract (MD+PDF), Git repository structure, Jira task screenshots, AI coding account evidence, Spec Kit self-learning documentation.

4.2.2 Sprint 2 (PA2) – 08 June 2026 to 21 June 2026

Sprint 2 Goal: Produce the initial Project Plan and Vision Document covering all 14 functional groups; initialize Spec Kit on GitHub; complete all meeting documentation and AI usage reporting.

	TASK	OUTCOME	WEEK 3 (JUN 8 - 14)							WEEK 4 (JUN 15-21)						
			08/06	09/06	10/06	11/06	12/06	13/06	14/06	15/06	16/06	17/06	18/06	19/06	20/06	21/06
MEETING	Sprint Planning Meeting	Planning report														
	Scrum Meeting #1	Meeting minutes														
	Scrum Meeting #2	Meeting minutes														
	Sprint Review Meeting	Review report														
DOCUMENTATION	Draft Project Plan (Intro, Overview, Org, Schedule)	Project Plan doc														
	Vision Doc – Intro, Positioning, Problem Statement	Vision doc sections														
	Vision Doc – Stakeholders, User Summary, Environment	Vision doc sections														
	Vision Doc – Product Features (all 14 FGs, 3–5 sent. each)	Feature descriptions														
	Vision Doc – Non-Functional Requirements	NFR section														
	Vision Doc – Workflow Diagrams: Donor Booking flow (Mermaid)	Workflow diagram														
	Vision Doc – Workflow Diagrams: SOS Emergency flow (Mermaid)	Workflow diagram														
	AI Usage Report	AI usage log														
SETUP	Spec Kit – Setup GitHub repo & run init commands	constitution.md + init files														
LEARNING	Spec Kit Training (YouTube tutorials) – All members	Training evidence/screenshots														
	HTML, CSS, JavaScript	Training evidence/screenshots														
	ReactJS, TailwindCSS, NodeJS	Training evidence/screenshots														

Figure 2: Sprint 2 Schedule

Task	Owner	Due Date
Project Plan – Intro, Project Overview	Kiệt (lead)	12/06/2026
Project Plan – Project Organization, Schedule, Build Plan	Kiệt + all members	12/06/2026
Vision Doc – Intro, Positioning, Problem Statement	Linh	12/06/2026
Vision Doc – Stakeholders, User Summary	Linh	14/06/2026
Vision Doc – Features: 1.1 User Account Mgmt, 1.2 Booking	Kiệt	16/06/2026
Vision Doc – Features: 1.3 Q&A AI, 1.4 News & Notifications	Quý	16/06/2026
Vision Doc – Features: 1.5 Donation Impact, 1.6 Community	Linh	16/06/2026
Vision Doc – Features: 2.1 Campaign Mgmt, 2.2 Communication, 2.3 Blood Inventory	Triết	16/06/2026
Vision Doc – Features: 3.1 SOS Hospital	Dương	16/06/2026
Vision Doc – Features: 4.1 User Automations, 4.2 BC Automations, 4.3 Notification Service	Triết	16/06/2026
Vision Doc – NFR + Workflow Diagrams (Mermaid)	Quý + Kiệt + Triết	18/06/2026
Spec Kit GitHub setup + constitution.md	All members individually	13/06/2026
AI Usage Report	Quý + All members	20/06/2026

Deliverables: PA2 submission – Project Plan (MD+PDF), Vision Document (MD+PDF), Spec Kit artifacts (constitution.md + init files), Meeting minutes, Jira screenshots, AI Usage Report, Git log.

4.2.3 Sprint 3 (PA3) – 22 June 2026 to 12 July 2026

Sprint 3 Goal: Deliver revised documents (Changes.md), complete use-case model and specifications with UI prototypes for all 14 functional groups, implement FG1 (1.1 User Account Management) end-to-end using Spec Kit, and produce v0.1 Alpha build.

TASK		OUTCOME	WEEK 5 (JUN 22 - 28)							WEEK 6 (JUN 29 - JUL 5)							WEEK 7 (JUL 6 - 12)						
			22/06	23/06	24/06	25/06	26/06	27/06	28/06	29/06	30/06	01/07	02/07	03/07	04/07	05/07	06/07	07/07	08/07	09/07	10/07	11/07	12/07
MEETING	Sprint Planning Meeting	Planning report																					
	Scrum Meeting #1	Meeting minutes																					
	Scrum Meeting #2	Meeting minutes																					
	Sprint Review Meeting	Review report																					
DOCUMENTATION	Revise Project Plan – 2nd submission + Changes.md	Updated plan + Changes.md																					
	Detailed Vision Doc – 2nd submission + Changes.md	Updated vision doc																					
	UC Diagrams – User Features (1.1–1.6) (Mermaid)	UC diagrams																					
	UC Diagrams – Blood Center (2.1–2.3) + Hospital (3.1) (Mermaid)	UC diagrams																					
	UC Diagrams – System Automations (4.1–4.3) (Mermaid)	UC diagrams																					
	UC Diagrams – Admin Features (5.1) (Mermaid)	UC diagrams																					
	UC Spec – 1.1 User Account Mgmt (Registration, OTP, Login, Profile)	UC spec + prototype																					
	UC Spec – 1.2 Donation Booking & Location	UC spec + prototype																					
	UC Spec – 1.3 Q&A AI (chatbot, eligibility, redirection)	UC spec + prototype																					
	UC Spec – 1.4 News, Notifications & SOS Alerts	UC spec + prototype																					
	UC Spec – 1.5 Donation Impact & Tracking (gamification)	UC spec + prototype																					
	UC Spec – 1.6 Community (Fairpage Redirect)	UC spec + prototype																					
	UC Spec – 2.1 Campaign & Donor Mgmt + QR scan	UC spec + prototype																					
	UC Spec – 2.2 Communication & Engagement Mgmt	UC spec + prototype																					
	UC Spec – 2.3 Blood Inventory & Emergency Coordination	UC spec + prototype																					
	UC Spec – 3.1 Emergency SOS Request (Hospital)	UC spec + prototype																					
	UC Spec – 4.1 User Automations + 4.2 BC Notification Service	UC spec + prototype																					
	UC Spec – 5.1 Administrator Features	UC spec + prototype																					
	AI Usage Report	Reports																					
	UI Prototypes for all 14 FGs (Figma/VoBot)	Prototype screenshots																					
DEVELOPMENT	FE	FE setup: React + Tailwind CSS project scaffold																					
	BE	BE setup: Node.js + Python API scaffold																					
	DB	DB setup: MongoDB schemas (User, Donation, Campaign, Blood Bag...)																					
VIDEO		Implement FG1 – 1.1 User Account Mgmt (full-stack, Spec Kit)																					
		Video demo – FG1 (YouTube Unlisted)																					

Figure 3: Sprint 3 Schedule

Task	Owner	Reviewer	Due Date
Revised Project Plan (Sec 1–4) + Changes.md	Kiệt	Linh	23/07/2026
Revised Vision Doc (all sections) + Changes.md	Kiệt	Linh	25/07/2026
UC Diagrams – User Features 1.1–1.6 (Mermaid)	Linh (1.1–1.3), Kiệt (1.4–1.5), Quý (1.6)	Triết	28/06/2026
UC Diagrams – Blood Center 2.1–2.3 + Hospital 3.1 (Mermaid)	Triết (2.1–2.3), Dương (3.1)	Linh	28/06/2026
UC Diagrams – System Automations 4.1–4.3 (Mermaid)	Triết	Dương	28/06/2026
UC Diagrams – Admin Features 5.1 (Mermaid)	Kiệt	Linh	28/06/2026
UC Spec – 1.1 User Account Mgmt (Registration, OTP, Login, History)	Kiệt	Linh	02/07/2026
UC Spec – 1.2 Donation Booking & Location Services	Kiệt	Linh	02/07/2026
UC Spec – 1.3 Q&A AI (chatbot, eligibility, pre/post guidance, redirection)	Quý	Linh	02/07/2026

Task	Owner	Reviewer	Due Date
UC Spec – 1.4 News, Notifications & SOS Emergency Alerts	Quý	Linh	02/07/2026
UC Spec – 1.5 Donation Impact & Tracking (Journey, gamification, badges)	Linh	Kiệt	03/07/2026
UC Spec – 1.6 Community (Fanpage Redirect, smart handling)	Linh	Kiệt	01/07/2026
UC Spec – 2.1 Campaign & Donor Mgmt (event creation, QR scan, screening)	Triết	Linh	03/07/2026
UC Spec – 2.2 Communication & Engagement Mgmt (CMS, broadcast)	Triết	Linh	03/07/2026
UC Spec – 2.3 Blood Inventory & Emergency Coordination	Triết	Linh	03/07/2026
UC Spec – 3.1 Emergency SOS Request Mgmt (Hospital)	Dương	Linh	03/07/2026
UC Spec – 4.1 User Automations + 4.2 BC Automations + 4.3 Notification Service	Triết	Linh	03/07/2026
UC Spec – 5.1 Admin Features	Kiệt	Linh	03/07/2026
UI Prototypes for all UCs (Figma/v0/Bolt) – Triết leads the FE design system; each member builds the prototype for the UCs they specced above	Triết (design system) + each member (own UCs)	Linh (UX consistency check)	07/07/2026
FG1 Implementation – 1.1 User Account Mgmt: Backend API + auth logic (Node.js)	Kiệt	Quý	07/07/2026
FG1 Implementation – 1.1 User Account Mgmt: Frontend UI + integration (React)	Triết	Quý	07/07/2026
FG1 Implementation – 1.1 User Account Mgmt: MongoDB schema (User, OTP, Session)	Dương	Quý	07/07/2026
Video demo FG1 (YouTube Unlisted, narrated)	Kiệt (compiles) + all members (narrate own part)	Linh	10/07/2026

Deliverables: PA3 submission – Changes.md, UC Model (MD+PDF), UC Specs + prototype screenshots for all 14 FGs (MD+PDF), Source code FG1 (no node_modules), Spec Kit

artifacts, YouTube demo link, Meeting minutes, Jira screenshots, AI Usage Report, Git log.

4.2.4 Sprint 4 (PA4) – 13 July 2026 to 02 August 2026

Sprint 4 Goal: Revise use-case specifications (all 14 FGs), produce complete software architecture documentation (C4 L1–L3, Deployment), and implement FG2 (1.2 Donation Booking & Location), FG3 (2.1 Blood Center Campaign & Donor Mgmt), FG5 (1.4 News, Notifications & SOS Emergency Alerts), FG8 (2.2 Communication & Engagement Mgmt), FG9 (2.3 Blood Inventory & Emergency Coordination), FG10 (3.1 Hospital SOS Request, Broadcast & Tracking), FG11 (4.1 User-Facing Automations), FG12 (4.2 Blood Center Automations), and FG13 (4.3 Notification Service) end-to-end using Spec Kit, and produce v0.2 Beta build.

TASK	OUTCOME	WEEK 8 (JUL 13 - 19)							WEEK 9 (JUL 20 - 26)							WEEK 10 (JUL 27 - AUG 2)						
		13/07	14/07	15/07	16/07	17/07	18/07	19/07	20/07	21/07	22/07	23/07	24/07	25/07	26/07	27/07	28/07	29/07	30/07	31/07	01/08	02/08
MEETING	Sprint Planning Meeting																					
	Scrum Meeting #1																					
	Scrum Meeting #2																					
	Sprint Review Meeting																					
DOCUMENTATION	Revised UC Spec – all 14 FGs, address TA feedback (Changes.md)																					
	Updated UC Specs + Changes.md																					
	Architecture Doc – Tech Stack description																					
	C4 L1 – System Context Diagram (Mermaid) + explanation																					
	C4 L2 – Container Diagram (Mermaid) + descriptions																					
	C4 L3 – Component Diagram: Frontend (Mermaid)																					
	C4 L3 – Component Diagram: Backend (Mermaid)																					
	Deployment Diagram (Mermaid) + node descriptions																					
	AI Usage Report																					
	Reports																					
DEVELOPMENT	Implement FG2 – 1.2 Donation Booking & Location (Spec Kit)																					
	Working code + Spec Kit artifacts																					
	Implement FG3 – 2.1 Blood Center Campaign & Donor Mgmt (Spec Kit)																					
	Working code + Spec Kit artifacts																					
	FG5 – 1.4 News, Notifications & SOS Emergency Alerts																					
	Working code + artifacts																					
	FG8 – 2.2 Communication & Engagement Mgmt																					
	Working code + artifacts																					
	FG11 – 4.1 User-Facing Automations																					
	Working code + artifacts																					
VIDEO	FG12 – 4.2 Blood Center Automations + FG13 – 4.3 Notification Service																					
	Working code + artifacts																					
	FG9 – 2.3 Blood Inventory & Emergency Coordination (Spec Kit)																					
	Working code + artifacts																					
	FG10 – 3.1 Hospital SOS Request, Broadcast & Tracking (Spec Kit)																					
	Working code + artifacts																					
TESTING	Spec Kit generated test cases – FG2 & FG3																					
	Test case files																					
VIDEO	Video demo – FG2 & FG3 (YouTube Unlisted)																					
	YouTube link																					

Figure 4: Sprint 4 Schedule

Task	Owner	Reviewer	Due Date
Revised UC Specs – all 14 FGs, address TA feedback (Changes.md)	Linh (lead, consolidates); each original spec owner revises their own section (see Sec 4.2.3 table for owners)	Kiệt	16/07/2026
Tech Stack description + C4 L1 System Context Diagram	Kiệt	Dương	18/07/2026
C4 L2 Container Diagram + descriptions	Dương (draft) + Kiệt (refine)	Triết	21/07/2026
C4 L3 Component Diagram – Frontend	Quý	Triết	23/07/2026
C4 L3 Component Diagram – Backend	Kiệt	Dương	23/07/2026

Task	Owner	Reviewer	Due Date
Deployment Diagram + node descriptions	Triết	Kiệt	24/07/2026
FG2 – 1.2 Donation Booking & Location: Backend API + Maps integration	Dương	Quý	30/07/2026
FG2 – 1.2 Donation Booking & Location: Frontend (map UI, booking flow, QR e-ticket)	Quý	Dương	30/07/2026
FG2 – 1.2 Donation Booking & Location: MongoDB schema (Booking, Location)	Dương	Quý	30/07/2026
FG3 – 2.1 Blood Center Campaign & Donor Mgmt: Backend (event, QR check-in, eligibility logic)	Triết	Linh	30/07/2026
FG3 – 2.1 Blood Center Campaign & Donor Mgmt: Frontend (campaign dashboard, check-in UI)	Linh	Triết	30/07/2026
FG3 – 2.1 Blood Center Campaign & Donor Mgmt: MongoDB schema (Campaign, Session)	Dương	Triết	30/07/2026
FG5 – 1.4 News, Notifications & SOS Emergency Alerts (push, email)	Kiệt (BE, lead) + Quý (FE) + Linh (FE)	Triết	22/07/2026
FG8 – 2.2 Communication & Engagement Mgmt (CMS, automated reminders, broadcast)	Triết (FE+BE, lead) + Kiệt (BE) + Linh (FE)	Quý	24/07/2026
FG9 – 2.3 Blood Inventory & Emergency Coordination (real-time tracking, auto SOS)	Kiệt (BE, lead) + Triết (BE) + Dương (DB) + Linh (FE)	Triết	24/07/2026
FG10 – 3.1 Hospital SOS Request Mgmt (create, broadcast, supply confirm & track)	Dương (BE, lead) + Quý (FE)	Kiệt	25/07/2026

Task	Owner	Reviewer	Due Date
FG11 – 4.1 User-Facing Automations (84-day rule, pre-screening form, e-ticket, SOS proximity alert)	Kiệt (BE, lead) + Triết (BE) + Quý (FE)	Dương	25/07/2026
FG12 – 4.2 BC-Facing Automations (digital donor record, lifecycle comms, shelf life, SOS prioritization)	Triết (BE, lead) + Kiệt (BE) + Quý (FE)	Dương	25/07/2026
FG13 – 4.3 Notification Service	Quý (lead) + Kiệt (BE)	Triết	25/07/2026
Spec Kit generated test cases – FG2 & FG3	Owner of each FG generates their own	Linh (QA lead consolidates & reviews)	29/07/2026
Video demo FG2 + FG3 (YouTube Unlisted)	Kiệt (compiles) + all members (narrate own part)	Linh	01/08/2026

Deliverables: PA4 submission – Changes.md, Architecture doc (C4 L1–L3 + Deployment, MD+PDF), Source code FG2+FG3+FG5+FG8+FG9+FG10+FG11+FG12+FG13, Spec Kit artifacts, generated test cases, YouTube demo link, Meeting minutes, Jira screenshots, AI Usage Report, Git log.

4.2.5 Sprint 5 (PA5) – 03 August 2026 to 23 August 2026

Sprint 5 Goal: Implement the remaining 4 functional groups, review and refine all Spec Kit test cases, write and execute test plan (≥50 test cases across ≥5 use cases including AI feature testing), produce bug reports, compile reflective report, and deliver the final product demo (15 minutes live).

	TASK	OUTCOME	WEEK 11 (AUG 3 - 9)							WEEK 12 (AUG 10 - 16)							WEEK 13 (AUG 17 - 23)						
			03/08	04/08	05/08	06/08	07/08	08/08	09/08	10/08	11/08	12/08	13/08	14/08	15/08	16/08	17/08	18/08	19/08	20/08	21/08	22/08	23/08
MEETING	Sprint Planning Meeting	Planning report																					
	Scrum Meeting #1	Meeting minutes																					
	Scrum Meeting #2	Meeting minutes																					
	Sprint Review Meeting	Review report																					
DEVELOPMENT	FG4 – 1.3 Q&A AI Chatbot	Working code + artifacts																					
	FG14 – 5.1 Administrator Features	Working code + artifacts																					
	FG6 – 1.5 Donation Impact & Tracking / Journey	Working code + artifacts																					
	FG7 – 1.6 Community Fanpage Redirect	Working code + artifacts																					
	Review & refine all Spec Kit generated test cases	Refined test cases																					
	Test Execution – manual functional testing	Test execution results																					
	Final full-stack integration & QA	Stable v1.0 build																					
	Bug Report	Bug report document																					
DOCUMENTATION	Write Test Cases (≥50, ≥5 use cases, incl. AI feature testing)	Test case document																					
	Test Plan document	Test plan																					
	Reflective Report – All members	Reflective report																					
	Final Submission package (update all PA1–PA5 docs)	PA5-GroupXX.zi p																					
DEMO	AI Usage Report (full project)	Complete AI log																					
	Final Product Demo (15 min live, all members)	Demo session																					

Figure 5: Sprint 5 Schedule

Task	Owner (lead named first)	Reviewer	Due Date
FG4 – 1.3 Q&A AI (conversational chatbot, eligibility, smart redirection)	Quý (AI+FE, lead) + Dương (BE/Python) + Linh (FE)	Kiệt	12/08/2026
FG6 – 1.5 Donation Impact & Tracking (statistics, badges, gamification, Journey)	Triết (FE, lead) + Dương (DB) + Linh (BE+FE)	Linh's part reviewed by Quý; Triết/Dương's part reviewed by Kiệt	13/08/2026
FG7 – 1.6 Community Fanpage Redirect (smart app-vs-browser redirect + error handling)	Triết	Quý	09/08/2026
FG14 – 5.1 Administrator Features	Triết (BE, lead) + Kiệt (BE) + Quý (FE)	Linh	15/08/2026
Test Plan document	Linh (lead)	Kiệt	10/08/2026
Test Cases (≥50 across ≥5 UCs), review & refine Spec Kit generated	Linh (leads QA); each FG's Owner supplies their own Spec Kit-generated cases	Kiệt	16/08/2026
Test Execution + Bug Report	Linh (leads execution); each FG Owner fixes bugs found in their own FG	Kiệt	17/08/2026
Reflective Report (all sections + individual reflection)	All members, individually, one subsection each	Linh (consolidates & checks consistency)	20/08/2026
Final submission package (update all PA1–PA5 docs)	Kiệt (lead)	Linh	22/08/2026
Final Product Demo (15 min live, all members present each ≥1 feature)	All members, each presenting the FG(s) they owned	— (live TA evaluation)	22/08/2026

Deliverables: PA5 submission – Complete source code (4 remaining FGs: FG4, FG6, FG7, FG14), all Spec Kit artifacts, Test Plan + Test Cases + Execution Results + Bug Report (MD+PDF), Reflective Report (MD+PDF), updated PA1–PA5 docs, AI Usage Report, Git log, live demo.

4.3 Build Plan

The following releases are planned across the 5 sprints. Tech stack: React + Tailwind CSS (Frontend), Node.js + Python (Backend), MongoDB (Database).

Sprint	Version	Type	Key Features / Deliverables
PA1	–	Documentation	Project Proposal, Existing App Survey, Team Contract, Tools Setup
PA2	–	Documentation	Project Plan, Vision Document (all 14 FGs), Spec Kit Initialization
PA3	v0.1 – Alpha	First Working Build	UC Model + Specs + Prototypes for 14 FGs. FG1: 1.1 User Account Mgmt (React FE + Node.js BE + MongoDB)
PA4	v0.2 – Beta	Extended Build	FG2: 1.2 Donation Booking & Location. FG3: 2.1 Campaign & Donor Mgmt. FG5: 1.4 News, Notifications & SOS Alerts. FG8: 2.2 Communication & Engagement Mgmt. FG9: 2.3 Blood Inventory & Emergency Coordination. FG10: 3.1 Hospital SOS Request Mgmt. FG11: 4.1 User-Facing Automations. FG12: 4.2 BC-Facing Automations. FG13: 4.3 Notification Service. Architecture docs (C4 L1–L3, Deployment). Generated test cases.
PA5	v1.0 – Final	Final Release	FG4, FG6, FG7, FG14: Q&A AI, Donation Impact & Tracking/Journey, Community Fanpage Redirect, Admin Features. Test plan, execution, bug report. Final demo.

Each build (PA3, PA4) is demonstrated via a YouTube (Unlisted) video demo with narration. The final v1.0 build is demonstrated live in the PA5 demo session.

4.4 Functional Groups Implementation Mapping

The table below maps all 14 LifeLine functional groups (across all 5 feature sections) to their implementation sprint and responsible members.

FG	Functional Group	Sprint	Lead(s)	Tech Focus
FG1	1.1 User Account Mgmt – Registration (CCCD QR), OTP verification, Login, Password recovery, Profile Mgmt, Donation History & e-Certificates	PA3	Kiệt, Triết, Dương	React, Node.js, MongoDB
FG2	1.2 Donation Booking & Location – Interactive Map, Location Discovery, Appointment Scheduling, QR e-Ticket generation, Booking Confirmation notification	PA4	Kiệt, Triết, Dương	React, Maps API, Node.js, MongoDB
FG3	2.1 Blood Center – Campaign & Donor Mgmt – Event creation, QR scan check-	PA4	Triết, Kiệt,	React, Node.js, MongoDB, QR

FG	Functional Group	Sprint	Lead(s)	Tech Focus
	in, Eligibility evaluation (medical screening), Real-time profile updates, Session summary		Dương	scan
FG4	1.3 Donor Guidance & Q&A AI – AI-powered FAQ, Donation eligibility assessment, Pre/post-donation guidance, Conversational chatbot (multi-turn), Smart feature redirection	PA5	Quý, Dương	React, Python (LLM API), MongoDB
FG5	1.4 News, Notifications & Communication – News feed, Campaign announcements, Automated reminders (84-day eligibility), Notification preference management, SOS emergency alerts	PA5	Kiệt, Quý	React, Node.js, push notification service
FG6	1.5 Donation Impact & Tracking (Journey) – Donation timeline, Personal contribution statistics, Milestone badges & achievements, Gamification progress tracking (donor levels, virtual trees)	PA5	Triết, Dương	React, Node.js, MongoDB
FG7	1.6 Community – Donor community info display, Direct Facebook Fanpage link, Smart redirect (app vs browser), Error handling	PA5	Triết	React (redirect logic only)
FG8	2.2 Blood Center – Communication & Engagement Mgmt – Content publishing (CMS), Emergency announcements, Automated lifecycle communications	PA5	Triết, Kiệt	React, Node.js, MongoDB
FG9	2.3 Blood Center – Blood Inventory & Emergency Coordination – Blood bag inventory tracking, Automated SOS processing (deduct stock, notify transport, <3 min)	PA5	Dương, Kiệt	Python, Node.js, MongoDB
FG10	3.1 Hospital – Emergency SOS Request Mgmt – SOS request creation & validation, Emergency alert broadcast (push/email), Supply confirmation & tracking	PA5	Dương, Quý	Python (AI priority), Node.js, React, MongoDB
FG11	4.1 User-Facing Automations – 84-day interval validation, Pre-donation screening form generation, E-Ticket & QR generation, SOS proximity alerting	PA5	Kiệt, Dương	Node.js, Python, MongoDB

FG	Functional Group	Sprint	Lead(s)	Tech Focus
FG12	4.2 Blood Center-Facing Automations – Digital donor record generation, Automated lifecycle communications, Shelf life monitoring & expiration alerts, SOS request evaluation & prioritization	PA5	Dương, Kiệt	Python, Node.js, MongoDB
FG13	4.3 Notification Service	PA5	Quý, Kiệt	Python, Node.js, MongoDB
FG14	5.1 Administrator Features - User account management, Role and permission management, System activity monitoring, System configuration management	PA5	Kiệt, Triết	Python, Node.js, MongoDB